Matrices: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Definition and types

Formulas

$A = [a_{ij}]_{m\times n}$, mn elements; N elements $\Rightarrow$ number of orders = divisors of N
$\operatorname{tr} A = \sum a_{ii}$; $|A| = 0 \Leftrightarrow$ singular

KEY POINTS

- Row $1 \times n$, column $m \times 1$, horizontal $m < n$, vertical $m > n$, square $m = n$
- Diagonal $a_{ij} = 0\,(i \neq j)$; scalar $= kI$; upper triangular $a_{ij} = 0\,(i > j)$; lower $a_{ij} = 0\,(i < j)$
- Equal matrices: same order and $a_{ij} = b_{ij}$

2. Addition, scalar multiple, trace

Formulas

$A \pm B = [a_{ij} \pm b_{ij}]$ (same order), $kA = [ka_{ij}]$
$\operatorname{tr}(A \pm B) = \operatorname{tr} A \pm \operatorname{tr} B$, $|kA| = k^n|A|$

KEY POINTS

- Addition is commutative and associative; additive inverse $-A$
- $|A + B| \neq |A| + |B|$ in general

3. Multiplication

Formulas

$A_{m\times n}B_{n\times p} = (AB)_{m\times p}$, $(AB)_{ij} = \sum_k a_{ik}b_{kj}$
$|AB| = |A||B|$, $AI = IA = A$

KEY POINTS

- AB needs columns of A = rows of B; $AB \neq BA$ in general
- Associative and distributive; $AB = O$ does not force $A = O$ or $B = O$
- Product of a row, a matrix and a column is a 1×1 matrix (number)

4. Commuting, identities and powers

Formulas

$AB = BA$ commute; $AB = -BA$ anti-commute

$$(A \pm B)^2 = A^2 \pm AB \pm BA + B^2$$

KEY POINTS

- Algebraic identities hold only if $AB = BA$; $(A + kI)^n$ always expands binomially
- Powers only for square matrices; $f(A)$ uses I for the constant term
- A commutes with A^n, A^{-1}, kI, polynomials in A

5. Characteristic equation

Formulas

Order 2: $A^2 - (\operatorname{tr} A)A + |A|I = O$
Order 3: $A^3 - (\operatorname{tr} A)A^2 + CA - |A|I = O$

KEY POINTS

- C = sum of principal 2×2 minors; roots sum $= \operatorname{tr} A$, product $= |A|$
- Use: reduce high powers, divide polynomial by $c(x)$, find $A^{-1} = \frac{1}{|A|}(\dots)$

STEPS

1. Find $\operatorname{tr} A$ and $|A|$ (and C for order 3).
2. Write $A^2 = pA - qI$.

3. Substitute repeatedly to lower the power.

6. Idempotent, periodic, involutory, nilpotent

Formulas

Idempotent $A^2 = A$: $A^n = A$, $(I + A)^n = I + (2^n - 1)A$
Involutory $A^2 = I$: $A^{-1} = A$, $(I + A)^n = 2^{n-1}(I + A)$
Nilpotent $A^m = O$: $(I + A)^n = I + nA + \binom{n}{2}A^2 + \cdots$

KEY POINTS

- Idempotent $|A| = 0$ or 1; involutory $|A| = \pm 1$; periodic $A^{k+1} = A$ (least k = period)
- Nilpotent index $\leq$ order; triangular with zero diagonal is nilpotent
- $\frac{1}{2}(I \pm A)$ idempotent when A is involutory

7. Transpose

Formulas

$(A^T)^T = A$, $(kA)^T = kA^T$, $(A \pm B)^T = A^T \pm B^T$
$(AB)^T = B^T A^T$, $(ABC)^T = C^T B^T A^T$, $|A^T| = |A|$

8. Symmetric and skew-symmetric

Formulas

Symmetric $A^T = A$; skew $A^T = -A$

$$A = \tfrac{1}{2}(A + A^T) + \tfrac{1}{2}(A - A^T)$$

KEY POINTS

- Skew: diagonal 0, $\mathrm{tr} = 0$, odd order $|A| = 0$; symmetric and skew $\Rightarrow O$
- $A + A^T$, AA^T, $A^T A$ symmetric; $A - A^T$ skew
- Skew: odd power skew, even power symmetric; symmetric A, B: $AB - BA$ is skew, AB symmetric iff $AB = BA$
- Free elements: symmetric $\frac{n(n+1)}{2}$, skew $\frac{n(n-1)}{2}$

9. Orthogonal and rotation matrices

Formulas

$AA^T = A^T A = I$, $A^{-1} = A^T$, $|A| = \pm 1$
$R(\alpha)R(\beta) = R(\alpha + \beta)$, $R(\theta)^n = R(n\theta)$

$$\mathrm{tr}(AA^T) = \sum a_{ij}^2$$

KEY POINTS

- Rows (columns) are unit vectors, mutually perpendicular
- $R(\theta)^{-1} = R(-\theta) = R(\theta)^T$

10. Adjoint

Formulas

$\mathrm{adj}\, A = [C_{ij}]^T$, $A \,\mathrm{adj}\, A = |A| I$
$|\mathrm{adj}\, A| = |A|^{n-1}$, $\mathrm{adj}(\mathrm{adj}\, A) = |A|^{n-2} A$
$|\mathrm{adj}(\mathrm{adj}\, A)| = |A|^{(n-1)^2}$, $\mathrm{adj}(kA) = k^{n-1}\,\mathrm{adj}\, A$
$\mathrm{adj}(AB) = \mathrm{adj}\, B\, \mathrm{adj}\, A$, $\mathrm{adj}\, A^T = (\mathrm{adj}\, A)^T$
Order 2: $\mathrm{adj} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

STEPS

1. Peel nested adj/scalars from the outside: $|\mathrm{adj}\, M| = |M|^{n-1}$ and $|kM| = k^n |M|$.

11. Inverse

Formulas

$A^{-1} = \dfrac{\text{adj } A}{|A|}$, exists iff $|A| \neq 0$
$|A^{-1}| = \frac{1}{|A|}$, $(A^{-1})^{-1} = A$, $(A^T)^{-1} = (A^{-1})^T$
$(AB)^{-1} = B^{-1}A^{-1}$, $(kA)^{-1} = \frac{1}{k}A^{-1}$, $A^{-n} = (A^{-1})^n$

KEY POINTS

- Inverse from characteristic equation: $A^2 - pA + qI = O \Rightarrow A^{-1} = \frac{1}{q}(pI - A)$
- Diagonal inverse is diagonal; symmetric inverse is symmetric; $A^2 = I \Rightarrow A^{-1} = A$
- $|A| = \pm 1$ with integer entries $\Rightarrow$ integer inverse

12. Cancellation and $AB = O$

KEY POINTS

- $AB = AC \Rightarrow B = C$ only for invertible A (pre-multiply by A^{-1})
- $AB = O$, A non-singular $\Rightarrow B = O$; both non-zero $\Rightarrow |A| = |B| = 0$
- Factor tricks: $(P^2 + Q^2)(P - Q) = O$ when $P^3 = Q^3$, $P^2Q = Q^2P$

Watch out

Cancelling a singular matrix or on the wrong side.

13. Elementary operations

STEPS

1. Rows: write $[A|I]$, use only row operations until $[I|A^{-1}]$.
2. Columns: write A over I, use only column operations until I over A^{-1}.
3. Equation form: $A = IA$ (rows, act on left A and pre-factor I) or $A = AI$ (columns).

Watch out

Mixing row and column operations in one calculation.

14. System of equations and rank

Formulas

$AX = B$: $|A| \neq 0 \Rightarrow X = A^{-1}B = \dfrac{(\text{adj } A)B}{|A|}$
$|A| = 0$: $(\text{adj } A)B \neq O \Rightarrow$ no solution
$AX = O$ has non-zero solution iff $|A| = 0$

KEY POINTS

- $|A| = 0$ and $(\text{adj } A)B = O$: infinite solutions or none (check)
- Rank = order of the largest non-singular sub-matrix = number of non-zero rows in echelon form
- Matrix given by $AX_i = Y_i$: $A[X_1X_2X_3] = [Y_1Y_2Y_3]$; $AX = \lambda X \Rightarrow A^nX = \lambda^n X$
- Matrix equations: $AX = B \Rightarrow X = A^{-1}B$, $XA = B \Rightarrow X = BA^{-1}$

15. Permutation matrix and quadratic form

Formulas

$P^TP = I$, $|P| = \pm 1$; PA permutes rows, AP permutes columns
$P^k = I$, $k =$ lcm of cycle lengths
$X^TAX = \sum a_{ij}x_ix_j$ ($= 0$ if A skew)

KEY POINTS

- Permutation matrices of order n: $n!$; trace = number of fixed positions

16. Powers, sums and counting

STEPS

1. A^n: find pattern of A^2, A^3, or $A = \lambda I + N$, or reduce with $A^2 = pA + qI$.
2. Rank one $A = uv^T$: $A^n = (v^T u)^{n-1} A$; row sums $1 \Rightarrow A^n \mathbf{1} = \mathbf{1}$.
3. Counting matrices: translate the condition (e.g. $\operatorname{tr} A^T A = \sum a_{ij}^2$), count cell by cell, use the complement for $|A| \neq 0$.

17. Common mistakes checklist

STEPS

1. Do not swap factors: $AB \neq BA$; transpose and inverse reverse the order.
2. $|kA| = k^n |A|$, not $k|A|$; $|A + B| \neq |A| + |B|$.
3. Use $(A \pm B)^2$ identities only when $AB = BA$.
4. Constant term of $f(A)$ is a multiple of I.
5. Check $|A| \neq 0$ before writing A^{-1}; do not cancel singular matrices.
6. $\operatorname{adj}(kA) = k^{n-1} \operatorname{adj} A$ (not k^n); use the order n carefully.
7. Do not mix row and column operations.